

MIG ALLEY



BDG Beta Version 0.80

August 9th, 2002

Foreword:

This is a modified version of Rowan's 'MiG Alley' (MA). It was created by the BDG, the 'Battle of Britain / MiG Alley Development Group', building on the code that Rowan published. The name 'BDG' was created before we knew that Rowan would release the MA code as well, so they got all the bold characters...

This is a beta version. That means that we encourage any feedback, both positive and negative to the public 'MiG Alley' forum at SimHQ. Should you have any suggestions relating to the new features, for example the new variables or default values, then you should also post to the thread(s) about the beta. If you have other wishes, you are of course free to post to other threads at SimHQ, but we make no promises and think we have heard most wishes. For a link to the SimHQ message boards, please see the 'Links' section.

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1 Installation:

First, install MiG Alley from your original CD. Patch it up to version 1.23 afterwards; please see the links section on where to get it. Now on to the BDG stuff:

Make a backup of all files you will overwrite first. If you have not had a BDG version yet, then the files overwritten are:

- ♦ mig.exe in the main directory, the
- ♦ keybd3d.bin in the KEYBOARD directory and the
- ♦ miglang.dll in the directory ENGLISH\TEXT

If you have already had a version and tweaked the settings file 'BDG.TXT', then backup that as well.

Install into the main directory. If you install into another directory (not recommended), then copy or move all files, including subdirectories, over into the main directory (where the mig.exe is).

We have not gotten any non-English versions yet.

To change values, like the BDG switches or the Field of View (FoV) angles, you have to edit the BDG.TXT file with an ASCII editor. For more information on the BDG.TXT see chapter 3.

2 BDG Features Overview:

2.1 Zoom

We have extended the zoom range and implemented some new ways to change the zoom ('Field of View', FoV). In game, **[SHIFT]-[F9]**, **[SHIFT]-[F10]**, **[SHIFT]-[F11]** chooses small, medium, and large field of view respectively (zoomed in, medium, zoomed out). The values for these can be chosen by the player. For example, when firing rockets, a smaller value might give you better aim. You can use the key **[SHIFT]-[F12]** to toggle between two FoVs (toggle the zoom). If you hold **[SHIFT]-[F8]** pressed, the view will get smaller till it hits the minimum and then get larger again till it hits the maximum etc. We would suggest you test these and find your favorite way of switching the zoom. Then, if you are not completely satisfied with the zoom values you can achieve, you can learn in this reference section what values you can change in what way.

2.2 Enemy Position Indicator

There is a new variable in the BDG.TXT that permits you to switch off the enemy position indicator. If this is not set, the indicator will not show up, independent of whether you disable or enable the enemy position indicator and the virtual artificial horizon in the options inside the game. Thus, you can independently switch the two features (EPI and virtual horizon) on and off. The variable is called `ENEMY_POSITION_INDICATOR`.

2.3 Reversed Outside Padlock

[ALT]-[F6] toggles the 'Outside Reversed Padlock' View, which is essentially the same as Outside Padlock, but now the target plane is in the foreground while your plane is in the background. You are free to zoom and rotate around the target plane, just like in normal padlock view. However, you can not target waypoints, and this view is also limited to a range of about 6 nautical miles. If you try to padlock a waypoint you are reset to outside track view.

2.4 Padlock Enemy in Center View

We have implemented a 'padlock center screen' function, which does, however, only work for enemy airplanes. By pressing the **[S]** key you can cycle through all enemies which are in a (by default) 20 by 20 degrees area around the center of the screen. If there are no enemies in this area, the message 'View not available' will appear at the bottom of the screen. You can change the area where enemies will be padlocked, please see the reference section.

2.5 Changes to the Map Screen

We hope to get more insight into the RTB-Bug by looking at the AI-Flight Movecodes, which can be displayed in the Map-Screen by activating the MA_DEBUG_MAP Boolean value. Once it is activated, you can toggle between displaying the flights' movecode and total amount of fuel/callsign by pressing **3** (that also enters the radio-commands menu which you can exit by pressing **0**). Be reminded of a feature called 'migcheat', which also displays MiG flights by pressing **CTRL**-**M**.

2.6 Ability to Invert View-Input

MiG Alley presents the possibility to select different devices to pan around while in cockpit or outside view, but as opposed to BoB it does not let you invert those inputs. We have implemented the INVERT_VIEW_PITCH variable which lets you do just that.

2.7 WinAmp Support

Since MiG Alley can only play (quite antique) midis, Retro added external WinAmp support. You can now control the WinAmp application from within MiG Alley. All you have to do is to start WinAmp prior to executing the mig.exe, and open a playlist. In the 3d, you can use the following commands:

- | | |
|--------------------------------------|--------------------------------------|
| ♦ Start Playing | CTRL - F10 |
| ♦ Stop Playing | CTRL - F11 |
| ♦ Jump to previous track in playlist | CTRL - F9 |
| ♦ Jump to previous track in playlist | CTRL - F12 |
| ♦ Increase volume | ALT - (Num) - + |
| ♦ Decrease volume | ALT - (Num) - - |

Please note that currently only playback in the 3d is supported, if you crash or exit the 3d otherwise, the music will fade out and stop. This feature should work with any reasonably recent WinAmp-version; download it for free at <http://www.winamp.com> .

2.8 Padlock to Infinity

This function is still at a very early stage and only implemented to give you a glimpse of what it should become... the idea is basically to padlock 'nothing at all' but instead stabilize the view in a certain direction. It should work as long as you are flying with pitch and roll close to zero (read: straight and level). You can activate this view by pressing **CTRL**-**S**, but please note that it is **very** (<- that is bold, underscored and italic !!) important to deactivate it again before changing waypoints or calling 'regroup' or 'disengage'.

2.9 Disabling Friendly Collisions

Quite important in Battle of Britain, a nice feature for MiG Alley is the option to deactivate collisions with friendly airplanes. You can do this by adding the `NO_FRIENDLY_COLLISIONS` in the `BDG.TXT`. Note that extensive flying inside friendly airplanes could lead to a noticeable slowdown caused by the collision-detection algorithm.

2.10 Changes to the Padlock

We have added additional information about the currently padlocked target, which may be accessed by activating the targeting diamond (`[D]`), and then pressing `[ALT]-[D]`. This will activate text next to the padlock box which shows (in the top line) your current airspeed, the difference between your target's and your speed, (in the second line) the relative altitude to your target and (in the third line) your target's speed. The speed information is always in knots, regardless of your 'speed units' selection in the preferences. In order to gain more information about MiG Alley's AI we have implemented the possibility to show your padlocked target's callsign and callnumber, and also its maneuver and movecodes. You can activate this function by pressing (1) `[D]` (activating the targeting diamond), then (2) `[ALT]-[D]` (activating the text next to the box), then (3) `[ALTGr]-[D]`. This procedure also ensures that only people who bring the necessary amount of masochism needed for testing can access this info.

Additional (optional) changes that were made to the padlock are a resizable targeting diamond, which color can also be changed from red to red-white. The text next to the box changes its color based on relative altitude, if the target is higher than you its color is black, if it is lower its color is white.

2.11 Continue / Quit Box

If you ever landed successfully in MiG Alley you may have noted that the game exits the 3d once your speed is below 0.1 knots. From BoB, we have ported a feature called 'Continue/Quit' box which pops up once your speed falls below 0.1 knots and asks if you want to remain in the 3d. Press `[1]` to continue parking, or `[2]` to quit for debriefing.

2.12 Disable Head Bobbing

Using the `BDG` setting `HEAD_BOBBING` permits you to switch on or off head bobbing, in other words, the movement of the head under g-forces.

2.13 Minor Stuff

- ♦ **Osram fixed a bug that prevented users with WinXP and (especially Force feedback) joysticks to play (err... fly)**
- ♦ **Glop fixed some array write errors. We cannot really say what effect these bugs in the code had.**
- ♦ **You can now change the size of the 'virtual artificial horizon'. You can also choose the size of the 'enemy position indicator' (EPI) in the upper, left-hand corner of your screen.**
- ♦ **You may choose a pan speed factor to speed up panning via coolie hat.**
- ♦ **Because the executable has been rebuilt, it has another internal date and normally would not read settings and cam-files created with the official version. Now there will only be a warning when loading cam files and you can choose to read the cam file.**
- ♦ **Until now joysticks were probably initialized unnecessarily often. Thanks to Dave from Rowan for his tip to do this. This should help with the 'slow Force Feedback joystick problem' and might even help with joystick related stuttering. The only drawback is that you cannot plug in/out USB joysticks during gameplay and expect to go on playing.**
- ♦ **We fixed a small bug with the whiteout-code.**

3 Guide to the BDG.TXT:

This section deals with the various values you encounter in the BDG.TXT file, which lets you customize MiG Alley to an extent. The values you encounter here are either floating point values (like '1.2'), Booleans (ON/OFF) or Degree-Values.

You can edit the BDG.TXT using a simple ASCII editor like notepad. You should see all possible commands already entered, so all you have to do is to change to values of the options you want to see altered.

If you enter an invalid value, the program uses the default values instead, most of the time those are the same as in the 'vanilla' Rowan 1.23 installation.

3.1 Degree-Values

Field of View (FoV) values

- FOV_SMALL, FOV_MEDIUM, FOV_LARGE are the values used for **[SHIFT]-[F9]**, **[SHIFT]-[F10]** and **[SHIFT]-[F11]**. FOV_TOGGLE_SMALL and FOV_TOGGLE_LARGE are the two FoV-values you can toggle using the **[SHIFT]-[F12]** key.

FOV_INITIAL is the FoV used when MiG enters the 3d. FOV_MAXIMAL and FOV_MINIMAL determine the maximum and minimum FoV that you can reach with **[CTRL]-(Num)-[+]/[-]** and with **[SHIFT]-[F8]**.

If you don't set these two variables, then the allowed zoom range will be FOV_SMALL to FOV_LARGE. The executable itself limits the zooms to the range 2 degrees to 155. If you increase these values, you should keep one thing in mind:

In order to 'compress' so much of the world onto your screen a fish eye effect is necessary. It is like an extreme camera lens that makes the nose in a portrait half as large as the whole head, or like in a lens in the door.

Many don't like this and actually many think this is a bug; actually it is not a bug, but a distortion that enables you to cram so much field of view into the monitor.

Padlock Center:

- Configure the area where enemies will be picked up by pressing the **[S]** key. You can enter different values for pitch and heading, using PC_HDG and PC_PITCH. The center of the area is always the center of the screen; if you, for example, enter 10 degrees for heading, the actual area where enemies will be picked up is 20 degrees in heading (10 degrees both to the left and right). Likewise for pitch.

It might be a good idea making the pitch-area larger than the heading, since in a turn-fight your enemies will most likely be in the same plane with you.

3.2 Floating Point Values

♦ EPI_RADIUS/EPI_Y_RADIUS

You can also choose the radius of the 'enemy position indicator' in the upper, left-hand corner of your screen.

It is possible to set the X- and Y- size independently; if you change the radius it seems you look more edge on it or more downward onto its circle. Typical values are 50/10, so a typical ratio is 5:1 and if you for example want to make it smaller but keep its general appearance, you might choose 30/6, which again has a ratio of 5:1.

• PAN_SPEED_FACTOR

You may choose a factor that speeds up panning via coolie hat. For example '2' means the panning will be twice as fast as in Rowan's exe.

• ART_HORIZON_SIZE

You can now change the size of the 'virtual artificial horizon'. A typical value is 50; higher values will make it larger.

• MA_DIAMOND_MULT

Using this variable you can change the size of the target diamond. '1' is the default value, values bigger than '3' are not recommended.

3.3 Boolean values

• ENEMY_POSITION_INDICATOR

Lets you switch on and off the enemy position indicator. If it is not set, the indicator will never show up, independent of the options inside the game. Thus you can independently switch the two features EPI and virtual horizon on and off.

• NO_FRIENDLY_COLLISIONS

If set to ON, this disables collisions with friendlies.

• MA_DEBUG_MAP

Displays all kinds of info on the in-flight-map () if set to ON.

• MA_INVERT_VIEW_PITCH

Inverts the pitch input of the device you selected for view-panning if set to ON.

• MA_COLORED_DIAMOND

If set to ON this switch enables a red-white targeting diamond.

4 Keyboard Reference:

Here is a list of all *new* keyboard commands. If you are an underemployed graphics artist, you are herewith invited to come up with a new keycard.

:-)

ALT - D	Enable Text next to the padlock diamond
ALTGr - D	Toggle between ACM info and Debug information
S	Padlock Enemy in center of the screen
CTRL - S	Padlock an Area
ALT - F6	Toggle Outside Reversed Padlock
SHIFT - F8	Decrease FoV till at its minimum, then increase
SHIFT - F9	Small FoV
SHIFT - F10	Medium FoV
SHIFT - F11	Large FoV
SHIFT - F12	Toggle FoV
CTRL - F9	Previous Track
CTRL - F10	Start Playing
CTRL - F11	Stop Playing
CTRL - F12	Next Track
ALT - (Num) - +	Music Volume up
ALT - (Num) - -	Music Volume down

5 Frequently Asked Questions:

- *How can I deactivate the intro-video ?*
Just start MiG with a '-NOINTRO' parameter in the command line.
- *Huh ?*
Right-click on your MA-Icon and select 'Properties'. In the 'target' line, add '-NOINTRO'. If your path contains spaces, wrap the path (NOT the parameter) into double quotas.
(e.g.: "C:\Program Files\MiG Alley\mig.exe" -NOINTRO)
- *Is there a no-CD 'crack' ?*
Not per se, you will have to do a bit by yourself:
Copy the 'Smacker' folder from the CD in the MA directory.
Using Notepad, open the 'roots.dir' file in you MA game folder and change every line that refers to your CD-drive to the MA directory.
- *When I enter the map (M) the game crashes...*
During the 1.23 patch installation let the tt2 window finish. You should see 'TT2 finished' displayed in the DOS-window.
- *After playing, my desktop is acting funny... multiple icons activated etc...*
Try pressing (both) SHIFT keys repeatedly. That should solve this annoying problem.
- *The WinAmp stuff is great !! Back then, what kind of music did these guys listen to ?*
No idea, but 'Rudolph, The Red Nosed Reindeer' was in 1950's top 40, so at least Retro's choice is clear !
- *Kinda short FAQ !*
Yes, help us making it more comprehensive by posting at the SimHQ MiG Alley boards !

6 Known Issues

- ***Game crashes when entering quick missions***
- ***Bridge Attack Slowdown Bug***
When you are attacking a bridge target using the 'Begin your Run' radio command and succeed in destroying the bridge while your wingmen are still bombing, the framerate drops considerably. You can bring it back up by issuing the 'Leave Area' command. Also, you could task only one flight.
- ***Deactivated 'enemy skill' setting***
We discovered that the 'enemy skill' setting in the 'game' preferences does not have any impact on gameplay. Enemy AI pilots will have the same skills as 'blue' pilots, so if you want to encounter rookies in quickmissions and campaigns, use a 'low' or 'medium' blue skill setting.
- ***Configurable Keyboard***
Some of you may have used the application controls.exe in the MiG main-directory... as we have added new commands, we don't expect this will work any longer. Some testing is welcome though...
- ***Hey ! This list is even shorten than the FAQ !***
That's actually a good thing, isn't it ? Of course there are more issues, please help us identifying them on SimHQ's MiG Alley Message Boards.

7 Credits:

These features were implemented by Lukas 'Retro' Friembichler:
Collision detection, 'infinity view', padlock additions, continue/quit box, WinAmp stuff, most of the bugs.

These features were implemented by Wolfram 'Osram' Kuss:
Deactivation of head bobbing, removed date check, pan speed factor, size of artificial horizon, disabling of artificial horizon and/or EPI and lots of other things.

Also, he set up and administered CVS and does 'coding support' like explaining things and writing code to get M\$FS airplanes into 3D Canvas for use by MA/BoB... the BDG would be at loss without him.

This manual:

By Osram and Retro, who also did the conversion to pdf format and created the keyboard font. Thanks to Kanajana from the SimHQ boards for gathering all links.

Apart from this beta, there have been several jobs done by coders:
Tailslide looked into a debug version for us coders.
Glop found and fixed some Array write bugs.

Bricklayer wrote a 'How to compile' guide for both BoB and MiG.

Thanks also to RSColonel_131st who set up the BDG website.

'Thank you' to SimHQ for hosting the forums we use to coordinate the work.

All this would not have been possible without Rowan who did something that only very few commercial software producers do, and set their brain child free.
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8 Changelog:

Changes from 0.70 to 0.80 (August 9th, 2002)

- ♦ We implemented '*version 1 bins*', which allow multiple bins to be used for one 3D object, which allows higher polygon counts and better quality. We are still working on the first high quality 3D object
- ♦ Improved error-output
- ♦ Killed a nasty F86 related CTD-bug
- ♦ Implemented WinAmp support
- ♦ Made Padlock Diamond color and size configurable

Changes from Rowan's version (Rowan number 1.23) to (unofficial) BDG version 0.70 (April 24th, 2002):

- ♦ WinXP and (especially FFB) joystick bugfix
- ♦ Added 'Padlock Center' Function
- ♦ Added 'Outside Reversed' Padlocked
- ♦ Implemented 'No Friendly Collisions' option
- ♦ First try at 'Padlock to Infinity' (blame KC23 ;-)
- ♦ Added Continue/Quit box
- ♦ Added Map-Debug functions
- ♦ Made some changes to the padlock
- ♦ Added ability to invert View Input
- ♦ Fixed array write bugs
- ♦ Size of virtual artificial horizon configurable
- ♦ Size of EPI configurable
- ♦ Configurable Head Bobbing
- ♦ Implemented pan speed factor
- ♦ Removed date check
- ♦ Added zoom stuff
- ♦ Implemented possibility of disabling the EPI
- ♦ Joystick reinitialization fix

9 The BDG:

Current members of the BoB Developer Group

Bader	Texture artist, tester
Bricklayer	Programmer
Eko	3D Modeling
JamesB	Tester
KC23	Spring Offensive guru, our contact to SimHQ
Lobo_mau	3D Modeling
Maususer	Texture artist
Old Dux	Texture artist
Osram	All-round-guru
Retro	Programmer
Viper-	Bin-file-hero, tool programming

10Links:

10.1 Official Resources

Empire Interactive's Official MiG Alley Page:

<http://www.empireinteractive.com/migalley/>

MiG Alley Official Patch (Version 1.23):

<http://www.empireinteractive.com/support/patches.html>

MiG Alley Source Code:

<http://www.simhq.com/simhq3/sims/downloads/>

Rowan Software's Official Homepage:

<http://www.rowansoftware.com/rowan/welcome.htm>

10.2 Community sites

AxA's Fly & Fight:

<http://www.flyandfight.com/>

BDG Web:

<http://www.3d-raumplan.com/Flightsim/>

Frugal's World of Simulations:

<http://www.frugalsworld.com/>

Frugal's World of Simulations Forums:

<http://forums.frugalsworld.com/vbb/>

Harry M's MiG Alley Annex:

<http://www.redshift.com/~harrym/migalley/migalley.html>

Hengist's MiG Alley Site:

<http://www.hengist.co.uk/MiGAlley/>

KC23's MiG Alley Strategy Guide:

<http://www.frugalsworld.com/migalley/index.shtml>

Maususer's MiG Alley Modifications

<http://www-personal.umich.edu/~mauser/>

Retro's MiG Alley downloads

<http://stud3.tuwien.ac.at/~e9826217/MA/index.html>

SimHQ.com:

<http://www.simhq.com/>

SimHQ Message Boards:

<http://www.simhq.com/cgi-bin/boards/cgi-bin/Ultimate.cgi>

Viper-´s Modifications:

<http://wbviper-.bei.t-online.de/vpbob.htm>

10.3 History

USAF Museum – Korean War Gallery

<http://www.wpafb.af.mil/museum/history/korea/korea.htm>

Korean War 50th Anniversary Home Page

<http://korea50.army.mil/>

If you find any bugs, you can keep them
...or notify us at the public 'MiG Alley' forum at SimHQ.